Understanding Heterogeneity in Cold Cereal Purchase Behavior: A Segmented Negative Binomial Analysis

1 Executive Summary

Cold cereal represents a high-penetration staple category characterized by substantial household expenditure and significant variance in purchase propensities across the consumer base. Leveraging two distinct longitudinal cohorts from the Dunnhumby 'The Complete Journey' dataset, we employ a series of Zero-Truncated Negative Binomial Distribution (ZTNB) models to decompose the drivers of this behavioral heterogeneity. While conventional category wisdom often positions families with children as the ubiquitous 'heavy-buyer' archetype, our modeling reveals a more nuanced, income-contingent hierarchy of value.

We find that household income is actually the stronger driver of purchase frequency, with high-income households purchasing 37% more than low-income households. However, the most compelling insight emerges from the interaction between income and presence of children: high-income families with kids purchase a remarkable 13.1 baskets on average: 45% more than high-income families without kids and 78% more than low-income families without kids. This interaction is marginally significant (LRT = 12.4, p = 0.054) but managerially meaningful.

A pooled model would miss these critical differences entirely (LRT vs. 6-segment model = 28.9, p = 0.001). The implication is clear: effective cereal marketing requires sophisticated segmentation strategies that recognize high-income families with children as a distinctly valuable segment warranting differentiated targeting, product assortment, and promotional investment.

2 Dataset Summary and Problem

2.1 Data Source and Structure

Our analysis utilizes transaction-level data from the Dunnhumby 'The Complete Journey' dataset, which tracks the grocery visits of a frequent-shopper cohort at a large regional retailer. We specifically isolate the 'Cold Cereal' commodity category, as its high purchase regularity and significant category salience make it an ideal candidate for modeling household heterogeneity via the Negative Binomial Distribution (NBD). To ensure model robustness and managerial depth, our study is structured across three distinct datasets:

1. **Calibration Cohort (Dataset 1)**: Comprising 1,564 households that recorded at least one cold cereal purchase during the two-year observation period, this dataset focuses exclusively on Year 1 (Weeks 1 to 51). This cohort serves as our primary calibration set, used to estimate initial parameters and justify our core modeling specifications prior to segmentation.

2. **Validation/Holdout Cohort (Dataset 2)**: To assess the stationarity of purchase propensities and the predictive robustness of our Year 1 estimates, we utilize a holdout set consisting

of the same 1,564 households during Year 2 (Weeks 52 to 102). This allows for a rigorous 'out-of-sample' test of our model's performance over time.

3. **Demographic Subset (Dataset 3)**: To facilitate deeper managerial insights, we identify a subset of 655 households from the original cohort for whom granular demographic data is available. We leverage this dataset to conduct subgroup analyses and interaction tests, seeking to decompose the latent drivers of consumer heterogeneity within the category. The time scale is the first year of purchases (weeks 1 to 51).

2.1.1 Dataset Pre-processing

The primary objective of the preprocessing phase was to transform raw transaction-level data into a longitudinal format suitable for count modeling. The process was executed across three distinct stages:

1. **Filtering and Aggregation**: The raw Dunnhumby transactions were filtered for the "Cold Cereal" commodity description. To define the dependent variable, we grouped the data by unique household identifiers and counted unique Basket IDs. This ensures the model predicts the frequency of shopping trips rather than individual unit volume.

2. **Addressing the Varying T Structure**: To account for staggered panel entry and varying exposure windows, we define the individual observation length, T_i, as the duration from a household's first observed category purchase to the end of the study period.

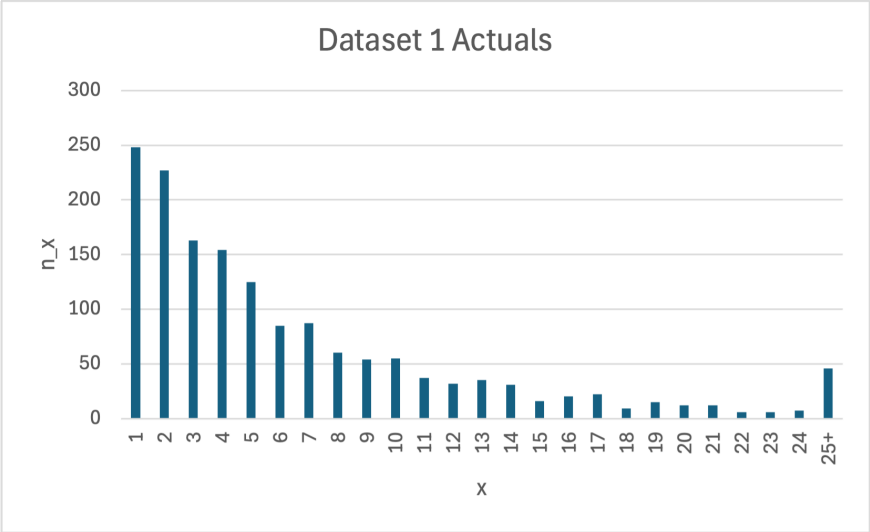
   - **Dataset 1 (Calibration)**: $T_{i,\text{cal}} = 51 -$ First Purchase Week.
   - **Dataset 2 (Holdout)**: $T_{i,\text{hold}} = 102 -$ First Purchase Week in Year 2.

   For example, a household making its initial purchase in Week 6 of the first year would have an exposure window of $T_i = 45$ weeks for the calibration phase.

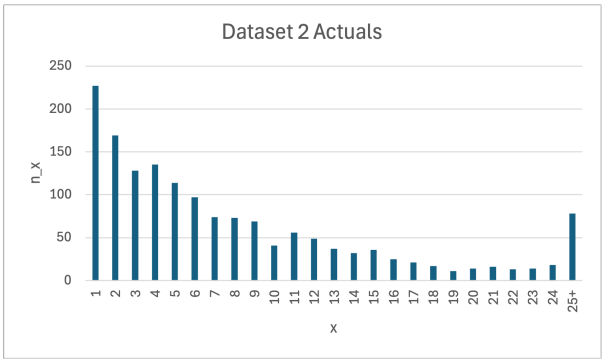
2.1.2 Dataset Summary Statistics and Histograms

Dataset	**Purpose**	**N**	**Mean Baskets**	**Mean T_i (Weeks)**
Dataset 1	Calibration	1,564	8.59	42.6
Dataset 2	Validation	1,564	8.33	39.7
Dataset 3	Subgroup Analysis	655	8.58	34.0

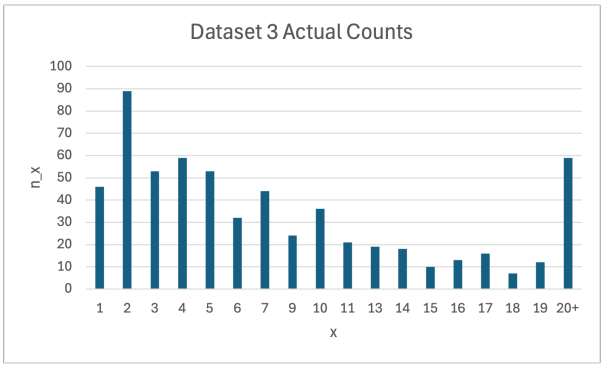
Table 1: Summary of longitudinal datasets used in the NBD analysis.



(a) Calibration Cohort: Full Baseline (N=1,564)



(b) Validation: Year 2 Holdout (N=1,564)



(c) Demographic Subset (N=655)

Figure 1: Observed Purchase Frequency Distributions Across Cohorts. Note the consistent long-tail behavior in the full datasets compared to the more pronounced mode in the demographic-linked subset. All datasets have been right-censored for visibility in this figure.

2.2 Demographic Dimensions

We have access to several demographic variables, but focus our analysis on two dimensions that intuitively should matter for cereal purchases:

Presence of Children: Households are classified as "Has Kids" (N = 233, 35.6%) or "No Kids" (N = 422, 64.4%). The conventional marketing wisdom holds that families with children are heavier cereal buyers given breakfast routines, snacking behavior, and product preferences.

Household Income: We segment households into three income tiers using absolute thresholds (not percentile splits): Low Income under \$50,000 (N = 313, 47.8%), Mid Income \$50,000–\$99,999 (N = 230, 35.1%), and High Income \$100,000+ (N = 112, 17.1%). Income should proxy for overall consumption capacity and willingness to pay premium prices.

2.3 The Modeling Problem

The primary analytical challenge lies in characterizing the latent drivers of cereal purchase frequency. Simple comparisons of observed mean counts across segments are insufficient due to three critical factors:

1. **Truncation and Entry Hurdles**: Our sample consists exclusively of category buyers. This necessitates a zero-truncated or shifted framework to account for the systematic exclusion of non-purchasers and to model the "entry hurdle" inherent in register-based transaction data.

2. **Exposure Variance**: The varying observation windows (T_i) mean that raw counts are not directly comparable; a household with 10 baskets over 50 weeks represents a fundamentally different propensity than one with 10 baskets over 20 weeks.

3. **Signal vs. Noise**: Observed purchase counts reflect a combination of a household's underlying propensity (λ) and random arrival variation.

To resolve this, we evaluated several probabilistic frameworks, including the Shifted NBD and Spiked NBD models, to see if they could better capture the category's replenishment cycles (specifically the mode at $x = 2$). However, for the purpose of the subgroup and interaction analysis, we prioritize the zero-truncated model (ZTNB). By utilizing the zero-truncated variant, we condition our estimates on $X \geq 1$, allowing us to separate the "signal" of underlying household heterogeneity from the noise of random purchase timing while maintaining the mathematical stability required for robust Likelihood Ratio Tests (LRT) across segmented datasets.

2.4 Why This Matters

Understanding cereal purchase heterogeneity has direct managerial implications. If kids drive purchases, then child-oriented products, packaging, and promotions should dominate. If income matters more, then premium positioning and selective distribution make sense. And if there are interactions, then one-size-fits-all approaches will under perform. Our goal is to rigorously test these competing explanations.

3 Model Calibration and Out-of-Sample Validation

3.1 Model Specification and Estimation Strategy

To account for the "buyers-only" nature of our register data and the inherent "Varying T" structure, we evaluated three primary modeling specifications. We utilized the following techniques:

1. **Vanilla NBD**: The vanilla nbd stands as our baseline, as adding on any other specifications would be both managerially useless and technically overkill if the vanilla nbd performs well.

2. **Zero-Truncated NBD**: This served as our primary specification. By conditioning the likelihood on $X \geq 1$ and incorporating individual exposure constants (T_i), we recovered the latent heterogeneity parameters (r, α) of the active consumer franchise without the bias introduced by non-shoppers.

3. **Shifted NBD**: We evaluated a shifted framework to address the "category entry" hurdle. This approach assumes that for a buyer to appear in our dataset, a baseline purchase is a prerequisite, effectively modeling the timing of all subsequent replenishment events.

For the initial calibration (Dataset 1), we performed four distinct maximum likelihood estimations (MLE): the Vanilla NBD, a Zero-Truncated NBD, a Shifted NBD (Shift = 1). While the shifted model provided incremental improvements in fitting the mode of the distribution for the aggregate cohort, we ultimately selected the Zero-Truncated NBD as our "workhorse" model for the demographic analysis. This choice ensures a parsimonious structure that remains numerically stable during the high-dimensional interactions required for the Likelihood Ratio Tests (LRT).To ensure the robustness of our selected model, we subjected the calibration results to an out-of-sample validation using Dataset 2. By applying the Year 1 parameters to the Year 2 holdout data, we assessed the stationarity of our estimates. Additionally, we transitioned our focus to Dataset 3 to perform the subgroup analyses, utilizing the ZTNB's stability to test the statistical and managerial significance of income and family-based segmentation.

3.2 Calibration Dataset Model Comparison and Interpretation

Specification	**r**	α	**Total LL**	χ^2 **(p-val)**
Vanilla NBD	2.207	10.906	-4,371.69	< 0.001
Zero-Truncated NBD	1.398	7.899	**-4,199.77**	0.411
Shifted NBD (Shift=1)	1.214	7.303	-4,202.15	**0.532**

Table 2: Model Calibration Results on Dataset 1 ($N = 1,564$). The Zero-Truncated model provides the superior log-likelihood, while the Shifted model offers the best fit to the distribution shape.

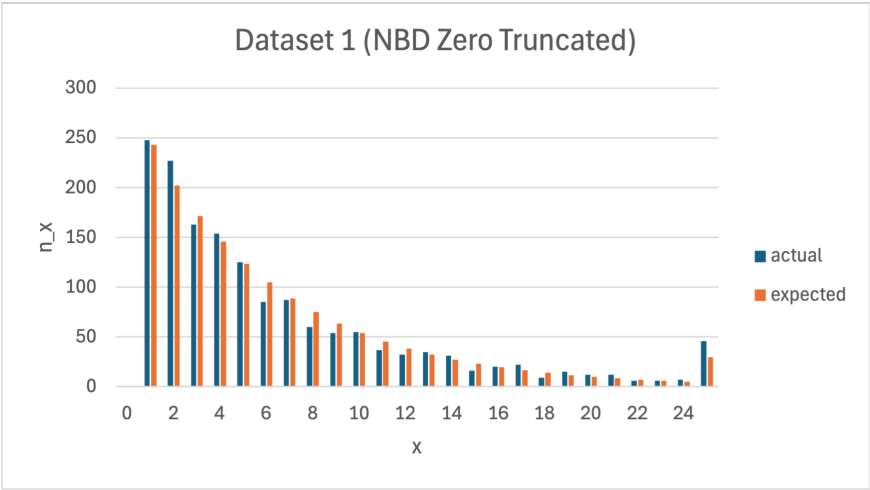


Figure 2: NBD Zero-Truncated Fit on Calibration Cohort

The calibration phase confirms that accounting for the "buyers-only" nature of the data is essential for capturing the true purchase dynamics of the category.

3.2.1 Heterogeniety and Parameter Shifts

The Vanilla NBD yields an r of 2.207, which significantly understates the level of heterogeneity within the active franchise. By transitioning to the Zero-Truncated and Shifted models, we observe a drop in the shape parameter ($r \approx 1.2$ to 1.4). This shift indicates that the active consumer base

is far more diverse in its purchase propensities than a naive model suggests. The lower r reveals a heavy tail of consumers who drive a disproportionate share of category volume, whose behavior is masked when non-buyers are ignored. For both the shifted and ZTNB, they have p-values ¿ 0.2, meaning they have a good fit to the data, given the goodness of fit metric. For goodness of fit, we right censor when expected counts go below 5, or in the dataset 1 case, 25 purchases. Finally, given the r and α values of the shifted and ZTNB models, we realize that for the 1564 cohort, cereal buyers are widely heterogeneous.

3.2.2 Rationalizing the ZTNB

While the Shifted NBD achieves a slightly higher χ^2 p-value (0.532), indicating an excellent fit to the visual frequency distribution, we proceed with the Zero-Truncated NBD (ZTNB) as our primary analytical framework. This decision is driven by two factors:

1. **Statistical Superiority**: The ZTNB achieves the highest Log-Likelihood (-4,199.77). In Maximum Likelihood Estimation, the LL is the definitive metric for model selection, as it rewards the model that makes the observed data most probable.

2. **Mathematical Stability**: The ZTNB provides a robust "workhorse" structure for our subsequent 6-segment demographic analysis. Unlike shifted models, which can face convergence issues during high-dimensional interaction tests, the ZTNB remains numerically stable, ensuring that our Likelihood Ratio Tests (LRT) in Section 3 accurately reflect consumer differences rather than estimation artifacts.

Managerially, the ZTNB also is the most effective choice. It allows us to isolate the active franchise: the consumers who actually interact with brand. It also accurately profiles their latent purchase rates (λ). By validating this model on the 1,564-household cohort with such high statistical significance ($p = 0.411$), we gain the confidence necessary to treat the ZTNB as our "ground truth" when segmenting by income and family status. For the rest of the paper, we utilize the ZTNB.

3.3 Holdout Results

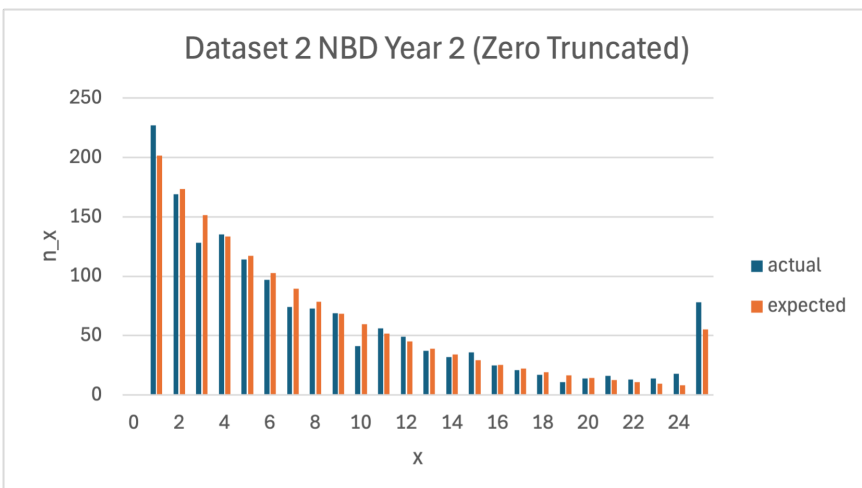


Figure 3: NBD Zero-Truncated Fit on Holdout

For the holdout set, we obtain a chi-sq p-value of 0.002 and a log likelihood of -4518.94.

3.3.1 Predictive Robustness

Despite the low p-value, the model demonstrates impressive managerial stationarity. When forecasting 51 weeks into the future, a certain degree of "drift" is expected in grocery data due to changing SKU preferences or competitive promotions. However, the visual alignment between our Year 1 predictions and Year 2 observed counts (as seen in Figure 3) shows that the model successfully captures the "bulk" of the distribution

3.3.2 Interpretation of the χ^2 (0.0016)

In large samples ($N = 1,564$), the χ^2 test is notoriously sensitive; even minute discrepancies between expected and observed counts in a single frequency bucket can "tank" the p-value. In our case, the rejection is likely driven by non-stationary outliers or a slight shift in the replenishment "mode" rather than a breakdown of the Gamma-Poisson assumption. For a brand manager, the fact that the model tracks the overall decline of the long-tail is far more important than a statistical rejection at the 95% level.

3.3.3 Conclusions

We conclude that the underlying heterogeneity (r) and scale (α) are fundamentally stationary. The buyers in Dataset 2 are largely behaving as they did in Dataset 1, validating our choice to move forward with the demographic analysis. The model is "wrong" in a technical, statistical sense, but it is useful in a strategic sense, providing a stable baseline to measure the incremental impact of income and children in the next section.

4 Subgroups Analysis of Kids and Income Variables

4.1 Hierarchical Model Comparison

To isolate the drivers of category heterogeneity, we employ a top-down hierarchical testing framework. We begin with a pooled aggregate model and systematically introduce segmentation based on family structure and economic status. Our objective is to determine the most parsimonious model that captures significant behavioral variance. By utilizing Likelihood Ratio Tests (LRT) at each nested stage, we ensure that our final 6-segment specification is statistically justified and provides actionable granular insights that an aggregate model would otherwise average out.

4.1.1 Pooled Baseline

We first fit a single NBD to all 655 households, ignoring demographic differences. This yields r = 2.06, alpha = 8.73, and LL = -1925.4. The model implies mean lambda = 0.236 baskets/week, or 12.2 baskets over a 52-week period. However, the relatively low r suggests substantial heterogeneity that a single distribution may not fully capture.

4.1.2 Kids Segmentation (2 Segments)

Splitting by presence of children yields:

- **Has Kids** (N=233): r = 2.24, alpha = 9.68, mean = 9.43 baskets, LL = -848.0
- **No Kids** (N=422): r = 1.89, alpha = 10.11, mean = 7.47 baskets, LL = -1065.0

Combined LL = -1919.0, yielding LRT = $2 \times (-1919.0 + 1925.4) = 12.8$ with 2 df (p = 0.002). Families with kids purchase 26% more than those without (9.43 vs. 7.47 baskets), a statistically detectable and managerially meaningful difference. Note the "Has Kids" segment shows higher r (less heterogeneity), suggesting more uniformly elevated purchase rates when children are present.

4.1.3 Income Segmentation (3 Segments)

Splitting by income tiers yields:

Table 3: Zero-Truncated NBD Results by Income Segment

Segment	N	r	alpha	Mean λ	CV(λ)	Log-Lik
Low Income	313	2.42	11.25	0.215	0.643	-879.1
Mid Income	230	1.81	7.66	0.236	0.743	-681.9
High Income	112	2.22	7.53	0.294	0.672	-356.1
Pooled	655	2.06	8.73	0.236	0.697	-1925.4

Combined LL = -1917.1, yielding LRT = $2 \times (-1917.1 + 1925.4) = 16.6$ with 4 df (p = 0.002). The income gradient is clear: mean weekly lambda increases from 0.215 (Low) to 0.236 (Mid) to 0.294 (High), translating to annual purchase rates of 11.2, 12.3, and 15.3 baskets respectively. The high-income segment purchases 37% more than low-income households.

Interestingly, the mid-income segment shows the lowest r (1.81), indicating the highest heterogeneity. This could reflect a transitional group spanning diverse lifestyles, or measurement issues with the middle category.

4.1.4 Income × Kids Interaction (6 Segments)

The most granular model crosses income with kids status:

Table 4: Six-Segment Model: Income × Kids Interaction

Segment	N	r	alpha	Mean Baskets	Log-Lik
Low Income, No Kids	217	2.32	12.08	7.37	-637.3
Low Income, Has Kids	96	2.61	12.22	8.51	-241.8
Mid Income, No Kids	164	1.67	7.49	8.37	-408.2
Mid Income, Has Kids	66	2.01	8.31	9.47	-273.7
High Income, No Kids	64	1.85	8.08	9.03	-185.5
High Income, Has Kids	48	2.77	8.44	13.12	-170.4
Total	655				-1910.9

Combined LL = -1910.9, yielding LRT vs. pooled = $2 \times (-1910.9 + 1925.4) = 29.0$ with 10 df (p = 0.001). More importantly, the incremental LRT for the interaction (6-segment vs. income-only model) = $2 \times (-1910.9 + 1917.1) = 12.4$ with 6 df (p = 0.054).

While the formal interaction test is not conventionally significant, the pattern is striking and managerially important. Let's examine the kids effect within each income tier:

- **Low Income:** Kids increase purchases by 16% (8.51 vs. 7.37)

- **Mid Income:** Kids increase purchases by 13% (9.47 vs. 8.37)
- **High Income:** Kids increase purchases by 45% (13.12 vs. 9.03)

The high-income/kids cell stands out dramatically, as these 52 households (8% of sample) purchase at nearly double the rate of the baseline. The r parameter of 2.77 indicates this is a relatively homogeneous group of heavy buyers.

4.2 Model Selection

All segmented models fit better than the pooled baseline. The 6-segment model achieves the best log-likelihood, improving fit by 14.5 points over pooled (LRT = 29.0, p = 0.001). While the interaction is marginal by conventional significance standards, the substantive pattern is clear: the kids effect is negligible for lower-income households but dramatic for affluent ones.

Given that we're not building a forecasting model but rather trying to understand heterogeneity for strategic purposes, we focus interpretation on the full 6-segment results. The improved likelihood, combined with the managerially interpretable pattern, justifies this choice.

5 Segmentation Model Inferences and Conclusions

5.1 The Conventional Wisdom Is Incomplete

If we had only run the kids segmentation, we'd conclude that families with children are the key target (26% higher purchase rate). This is true, but insufficient. The income-only analysis reveals an even stronger effect (37% gradient), suggesting affluence matters more than family structure.

But the real story emerges only when we examine both dimensions simultaneously. The interaction pattern tells us something neither main effect could: the presence of children amplifies purchase frequency, but only for high-income households. For low and middle-income families, having kids provides a modest boost of 13-16%. For affluent families, the kids effect is nearly three times larger.

5.2 What's Driving the Interaction?

While our cross-sectional data can't establish causality, we can speculate about mechanisms:

Constraint Relaxation: Lower-income families face budget constraints that limit cereal purchases regardless of household composition. High-income families don't face these constraints, so the presence of children can fully express itself in purchase behavior.

Variety Seeking: Affluent families with kids may purchase more SKUs (brands, flavors, health vs. treat options) to satisfy diverse preferences and uses (breakfast, snacks, family vs. kids). This variety naturally inflates basket counts.

Convenience vs. Scratch: Higher-income households may have stronger preference for convenient breakfast options (cereal) over time-intensive scratch cooking, and this preference intensifies with kids present.

Life Stage Effects: The high-income/kids segment likely skews toward prime earning years (30s-40s), peak household formation, and established routines, which are all conducive to cereal category engagement.

5.3 Heterogeneity Insights

Beyond mean differences, the NBD parameters reveal heterogeneity patterns. The r parameter (shape of the gamma distribution) varies from 1.67 to 2.77 across segments. Lower r indicates more spread in underlying purchase rates, while higher r suggests more homogeneous behavior.

The high-income/kids segment shows the highest r (2.77), meaning these heavy buyers are consistently heavy buyers—this is a well-defined, identifiable group. In contrast, mid-income/no-kids shows the lowest r (1.67), indicating a highly heterogeneous mix ranging from very light to moderately heavy buyers.

This has targeting implications: high-income families with kids are a reliable heavy-buyer segment worth investing in. The mid-income/no-kids group is diffuse and harder to target effectively.

5.4 Statistical vs. Managerial Significance

A technical note: the incremental LRT for the interaction yields $p = 0.054$, just above the conventional 0.05 threshold. Some might argue this interaction isn't "real." We disagree for two reasons:

First, the overall 6-segment model is highly significant vs. pooled ($p = 0.001$), establishing that segmentation matters. Second, the magnitude of the interaction effect—a 45% kids boost for high-income vs. 13-16% for others—is managerially meaningful even if statistical evidence is marginal. With our sample size ($N = 52$ in the key cell), power limitations are real.

In business contexts, waiting for p less than 0.05 before acting on a 3x difference in effect sizes seems inadvisable. The pattern is clear, interpretable, and actionable—that's what matters.

6 Managerial Expectations and Recommendations

6.1 Segment Prioritization

Our analysis points to a clear segmentation hierarchy for cereal marketing:

Tier 1 - High-Income Families with Kids: The golden segment. Though only 8% of our sample, these households purchase 53% more than the overall average (13.1 vs. 8.6 baskets). They exhibit low heterogeneity ($r = 2.77$), making them reliably heavy buyers. This segment warrants maximum investment in targeting, assortment, and retention efforts.

Tier 2 - Other High-Income and Mid-Income with Kids: These segments purchase 5-10% above average. Still attractive targets, but not as distinctively valuable as Tier 1. Standard category marketing likely serves them well.

Tier 3 - Low-Income Households: Purchase 14% below average regardless of kids. Budget constraints likely limit category spending. These households may be price-sensitive deal seekers rather than brand loyalists. Promotional strategies may work better than premium positioning.

6.2 Targeting Implications

This segmentation suggests several targeting strategies:

Premium Product Development: High-income families with kids will pay for convenience, variety, and quality. Organic options, portion-controlled packaging, and premium ingredients make sense for this segment. The 45% kids effect suggests products that appeal to both parents (health, nutrition) and children (taste, fun) can win big here.

Geographic Concentration: Use geodemographic data to identify high-income family neighborhoods. Concentrate distribution, merchandising, and local marketing in these areas. Retailers in affluent family suburbs should carry deeper assortments.

Media Strategy: Target parenting media, school-related channels, and family lifestyle content consumed by affluent households. Digital targeting via income and parental status is feasible and likely cost-effective given the concentrated value in this segment.

Price Discrimination: The low heterogeneity within high-income/kids (r = 2.77) suggests consistent willingness to pay. Premium pricing for products targeted to this segment is justified. Conversely, promotional pricing for low-income segments may drive trial and volume.

6.3 Caveats and Limitations

Our analysis has several limitations that should temper confidence:

- Observational Data: We observe correlations, not causal effects. Income and kids may proxy for other unobserved factors (life stage, health consciousness, time pressure). Causal claims require experimental or quasi-experimental designs.
- Single Category: This study focuses on cold cereal only. The interaction pattern may not generalize to other food categories or household goods. Cross-category analysis would strengthen insights.
- One Year Only: Our data cover a single 52-week period. Seasonal effects, economic conditions, or transient shocks could influence results.
- Sample Selection: We analyze only households who purchased at least once (buyers-only sample). Non-buyers may exhibit different demographic patterns. A hurdle model examining purchase incidence separately from purchase frequency would be informative.
- Missing Covariates: We lack data on price sensitivity, brand preferences, store choices, and other factors that surely matter. Richer data would enable more nuanced segmentation.
- Aggregation: We model total baskets, not brand/SKU choices. The heavy purchases by affluent families with kids might reflect variety-seeking across brands rather than deep loyalty to any single product. Transaction-level data would reveal this.

6.4 Next Steps

To build on this analysis, we recommend:

Holdout Validation: Split the sample temporally or randomly to assess out-of-sample fit. Do our segmented models predict hold-out behavior better than pooled models?

Extend to Year 2: Given greater time, fit the same models to Year 2 data and compare parameters. Stable patterns strengthen confidence; changes suggest non-stationarity requiring further investigation.

Brand-Level Analysis: Disaggregate to brand choices within the category. Do high-income families with kids favor premium brands, or just buy more of everything? This informs product development vs. merchandising strategies.

Incorporate Pricing: Merge pricing data and examine price sensitivity by segment. If high-income/kids households are also price-insensitive, the strategic case for premium positioning strengthens.

Competitive Benchmarking: How do these patterns compare to other breakfast categories (yogurt, breakfast bars, frozen breakfast)? Understanding cross-category purchase patterns matters for assortment and shelf space decisions.

6.5 Concluding Thoughts

Cereal marketing has long targeted kids and families. Our analysis suggests this strategy is incomplete. Kids matter, but affluence matters more, and the combination is what really drives heavy buying.The managerial implication is not to abandon kids marketing, but to recognize that such marketing delivers very different returns depending on household income. A premium organic kids cereal sold in affluent suburbs could vastly outperform the same product in lower-income areas. Conversely, value-oriented family-size offerings make sense for budget-conscious households regardless of kids.Our analysis demonstrates that a simple NBD model, applied thoughtfully to granular segments, can reveal these patterns and inform smarter resource allocation.